

Spring Framework



Member

Member

MemebeVo

- member package 생성
- MemberVo Class 생성

```
public class MemberVo {
    private int m_no;
    private String m_id;
    private String m_pw;
    private String m_name;
    private String m_gender;
    private String m_phone;
    private String m_email;
    private String m_role;
    private String m_reg_date;
    private String m_mod_date;

    public int getM_no() { return m_no; }
    public void setM_no(int m_no) { this.m_no = m_no; }
}
```

MemberRowMapper

- MemberRowMapper 생성

```
public class MemberRowMapper implements RowMapper<MemberVo> {

    @Override
    public MemberVo mapRow(ResultSet rs, int rowNum) throws SQLException {
        MemberVo member = new MemberVo();
        member.setM_no(rs.getInt("m_no"));
        member.setM_id(rs.getString("m_id"));
        member.setM_pw(rs.getString("m_pw"));
        member.setM_name(rs.getString("m_name"));
        member.setM_gender(rs.getString("m_gender"));
        member.setM_phone(rs.getString("m_phone"));
        member.setM_email(rs.getString("m_email"));
        member.setM_role(rs.getString("m_role"));
        return member;
    }
}
```

MemberDao

- MemberDao 생성 후 @Repository 어노테이션 추가

```
@Repository
public class MemberDao {

    private JdbcTemplate jdbcTemplate;

    @Autowired
    public MemberDao(JdbcTemplate jdbcTemplate) {
        this.jdbcTemplate = jdbcTemplate;
    }

    public void insertMember(MemberVo memberVo) {
        String sql = "INSERT INTO member(m_id, m_pw, m_name, m_gender,"
                + " m_phone, m_email, m_role, m_reg_date, m_mod_date)"
                + " VALUES(?,?,?,?,?,?,'USER',now(),now())";
        jdbcTemplate.update(sql, memberVo.getM_id(), memberVo.getM_pw(),
                memberVo.getM_name(), memberVo.getM_gender(),
                memberVo.getM_phone(), memberVo.getM_email());
    }
```

MemberDao

- MemberDao 생성 후 @Repository 어노테이션 추가

```
public MemberVo getDetailMember(int m_no) {
    String sql = "SELECT * FROM member WHERE m_no = ?";
    return jdbcTemplate.queryForObject(sql, new MemberRowMapper(), m_no);
}


public void updateMember(MemberVo memberVo) {
    String sql = "UPDATE member SET m_pw=?, m_name=?, m_gender=?,"
            + " m_phone=?, m_email=?, m_mod_date=now() WHERE m_no=?";
    jdbcTemplate.update(sql, memberVo.getM_pw(), memberVo.getM_name(),
            memberVo.getM_gender(), memberVo.getM_phone(),
            memberVo.getM_email(), memberVo.getM_no());
}
```

MemberController

- MemberController 생성 후 @Controller 어노테이션 추가

```
@Controller
public class MemberController {

    @Autowired
    private MemberDao memberDao;

    @RequestMapping("/registermember")
    public String registerMember() {
        return "user/register_member";
    }

    @RequestMapping("/registermemberconfirm")
    public String registerMemberConfirm(MemberVo memberVo) {
        memberDao.insertMember(memberVo);
        return "redirect:/loginmember";
    }
```

MemberController

- MemberController 생성 후 @Controller 어노테이션 추가

```
@RequestMapping("/modifymember")
public String modifyMember(@RequestParam("m_no") int m_no, Model model) {
    model.addAttribute("member", memberDao.getDetailMember(m_no));
    return "user/modify_member";
}

@RequestMapping("/modifymemberconfirm")
public String modifyMemberConfirm(MemberVo memberVo) {
    if (memberVo.getM_pw() == null || memberVo.getM_pw().isEmpty()) {
        MemberVo oldMember = memberDao.getDetailMember(memberVo.getM_no());
        memberVo.setM_pw(oldMember.getM_pw());
    }
    memberDao.updateMember(memberVo);
    return "redirect:/booklist";
}
```

아이디 중복확인

(fetch 비동기제어)

MemberController

- MemberController checkid mapping 추가
- Javascript alert 메시지 표시 @ResponseBody 어노테이션 추가

```
@RequestMapping("/checkid")
@ResponseBody
public String checkId(@RequestParam("m_id") String m_id) {
    return String.valueOf(memberDao.countById(m_id));
}
```

Member

MemberDao

```
public int countById(String m_id) {
    String sql = "SELECT count(*) FROM member WHERE m_id = ?";
    return jdbcTemplate.queryForObject(sql, Integer.class, m_id);
}
```

register_member.jsp

- 아이디 입력란 input 태그에 id 추가
- 버튼 태그에 id 추가
- </form> 뒤에 <script> 추가

```
<div class="row mb-3">
    <label class="col-sm-3 col-form-label">아이디</label>
    <div class="col-sm-9 d-flex gap-2">
        <input type="text" name="m_id" id="m_id" class="form-control">
        <button type="button" id="checkIdBtn"
            class="btn btn-warning text-white flex-shrink-0">중복확인</button>
    </div>
</div>
<script>
    let idChecked = false;

    document.getElementById("checkIdBtn").addEventListener("click", function () {
        const m_id = document.getElementById("m_id").value.trim();
        if (m_id === "") {
            alert("아이디를 입력하세요.");
            return;
        }
```

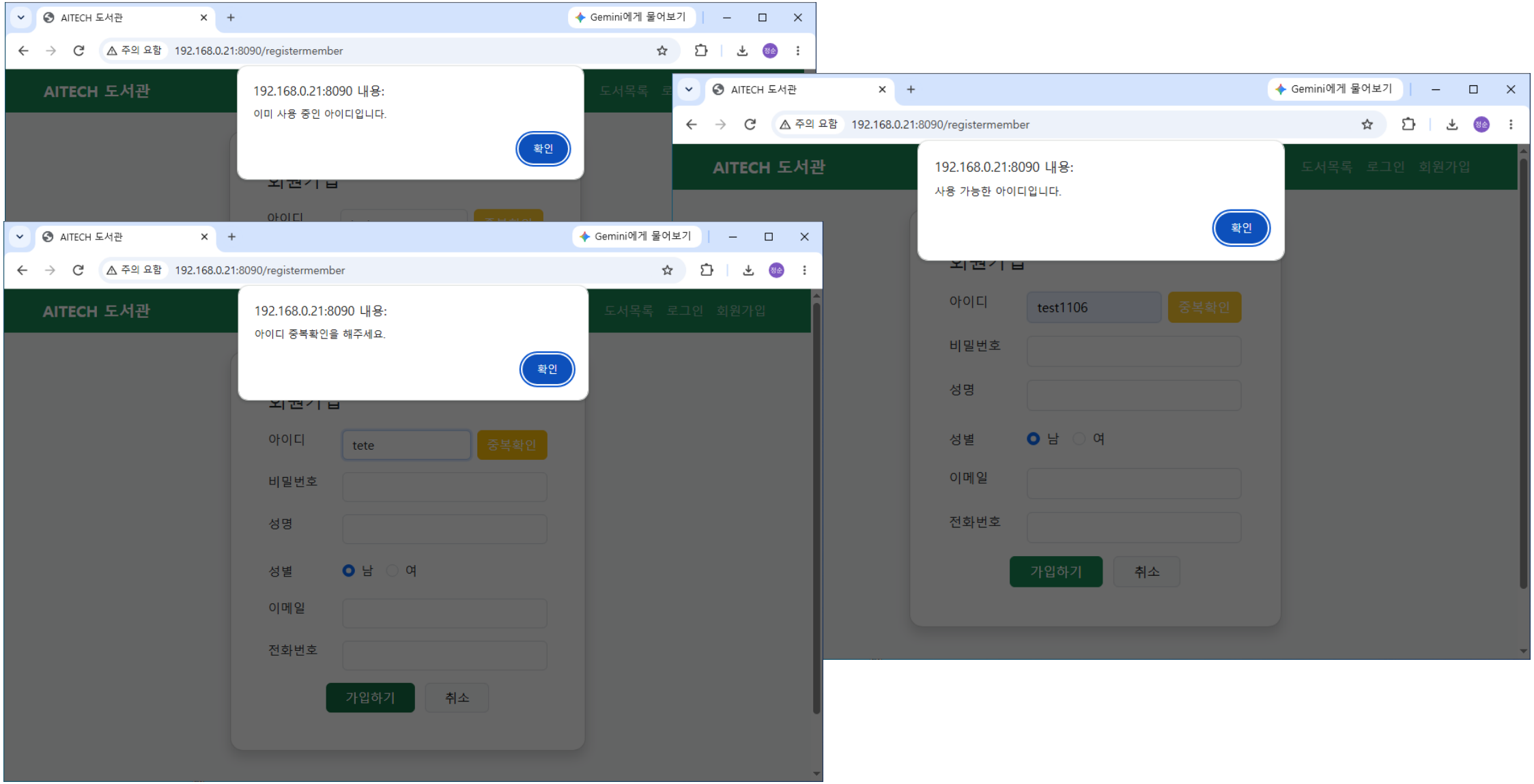
register_member.jsp

```
        fetch("checkid?m_id=" + m_id)
            .then(res -> res.text())
            .then(count => {
                if (count == 0) {
                    alert("사용 가능한 아이디입니다.");
                    idChecked = true;
                } else {
                    alert("이미 사용 중인 아이디입니다.");
                    idChecked = false;
                }
            });
    });

    document.getElementById("m_id").addEventListener("input", function () {
        idChecked = false;
    });

    document.querySelector("form").addEventListener("submit", function (e) {
        if (!idChecked) {
            e.preventDefault();
            alert("아이디 중복확인을 해주세요.");
        }
    });
</script>
```

참고화면


AITECH 도서관
Gemini에게 물어보기
주의 요함
192.168.0.21:8090/registermember
192.168.0.21:8090 내용:
이미 사용 중인 아이디입니다.
확인
AITECH 도서관
도서목록 로그인 회원가입
192.168.0.21:8090 내용:
사용 가능한 아이디입니다.
확인
아이디
test1106
중복확인
비밀번호
성명
성별
남
여
이메일
전화번호
가입하기
취소
AITECH 도서관
192.168.0.21:8090 내용:
아이디 중복확인을 해주세요.
확인
아이디
tete
중복확인
비밀번호
성명
성별
남
여
이메일
전화번호
가입하기
취소

로그인 · 로그아웃 세션

MemberController

- HttpSession 추가

```
@RequestMapping("/loginmember")
public String loginMember() {
    return "user/login_member";
}

@RequestMapping("/loginmemberconfirm")
public String loginMemberConfirm(@RequestParam("m_id") String m_id,
        @RequestParam("m_pw") String m_pw, HttpSession session) {
    MemberVo member = memberDao.getMemberById(m_id);
    if (member != null && member.getM_pw().equals(m_pw)) {
        session.setAttribute("loginMember", member);
        return "redirect:/booklist";
    }
    return "redirect:/loginmember";
}
```

MemberController

- HttpSession 추가

```
@RequestMapping("/logoutmember")
public String logoutMember(HttpSession session) {
    session.invalidate();
    return "redirect:/booklist";
}
```

MemberDao

```
public MemberVo getMemberById(String m_id) {
    String sql = "SELECT * FROM member WHERE m_id = ?";
    try {
        return jdbcTemplate.queryForObject(sql, new MemberRowMapper(), m_id);
    } catch (EmptyResultDataAccessException e) {
        return null;
    }
}
```

login_member.jsp

- <form> 태그 추가

```
<form method="post" action="loginmemberconfirm">

    <label class="form-label">아이디</label>
    <input type="text" name="m_id" class="form-control mb-3" placeholder="아이디">

    <label class="form-label">비밀번호</label>
    <input type="password" name="m_pw" class="form-control mb-4" placeholder="비밀번호">

    <div class="d-grid gap-2">
        <button type="submit" class="btn btn-success">로그인</button>
    </div>

    <p class="text-center mt-4 mb-0">
        <a href="registermember" class="text-decoration-none fw-semibold">회원가입</a>
        <a href="findpassword" class="text-decoration-none fw-semibold ms-3">비밀번호 찾기</a>
    </p>

</form>
```

header.jsp

- jstl 태그 추가

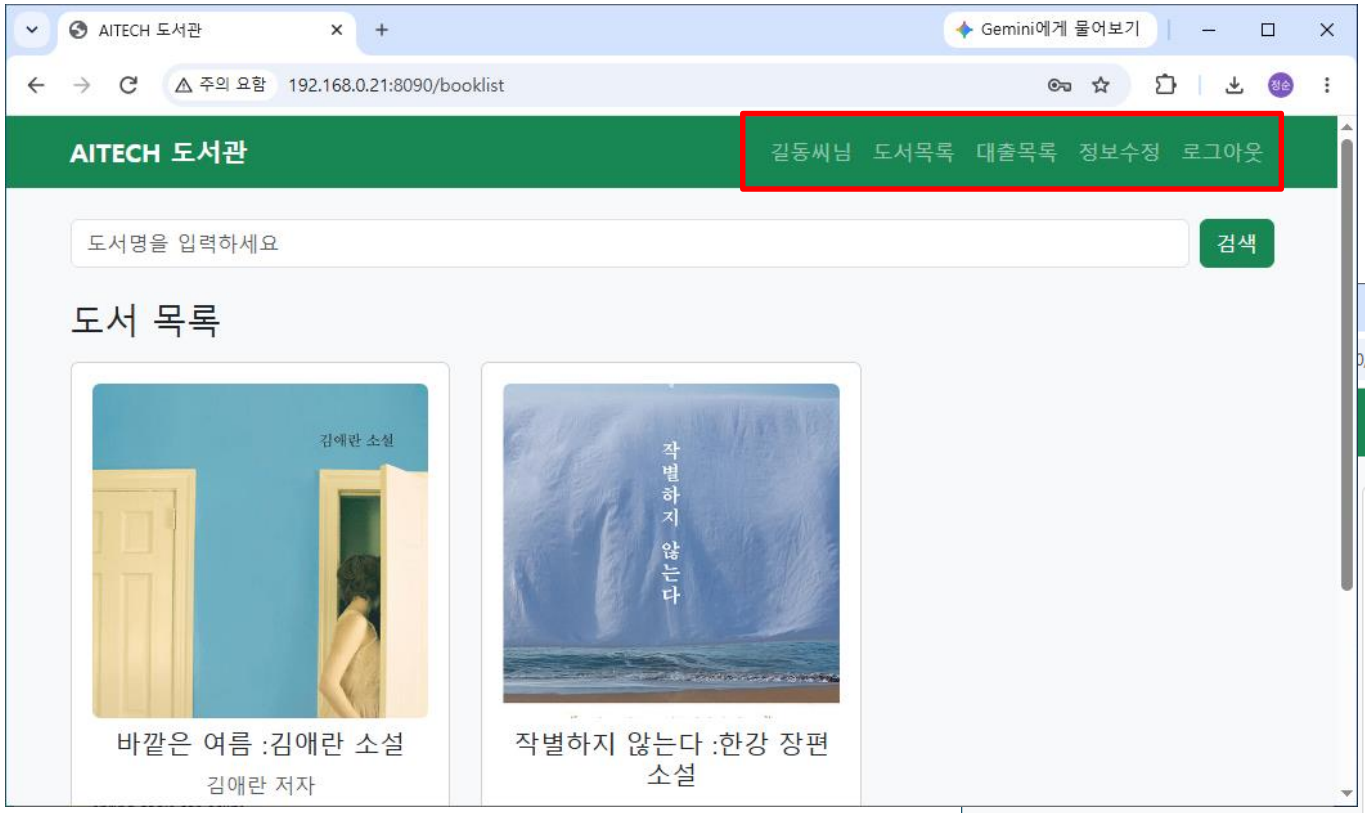
```
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

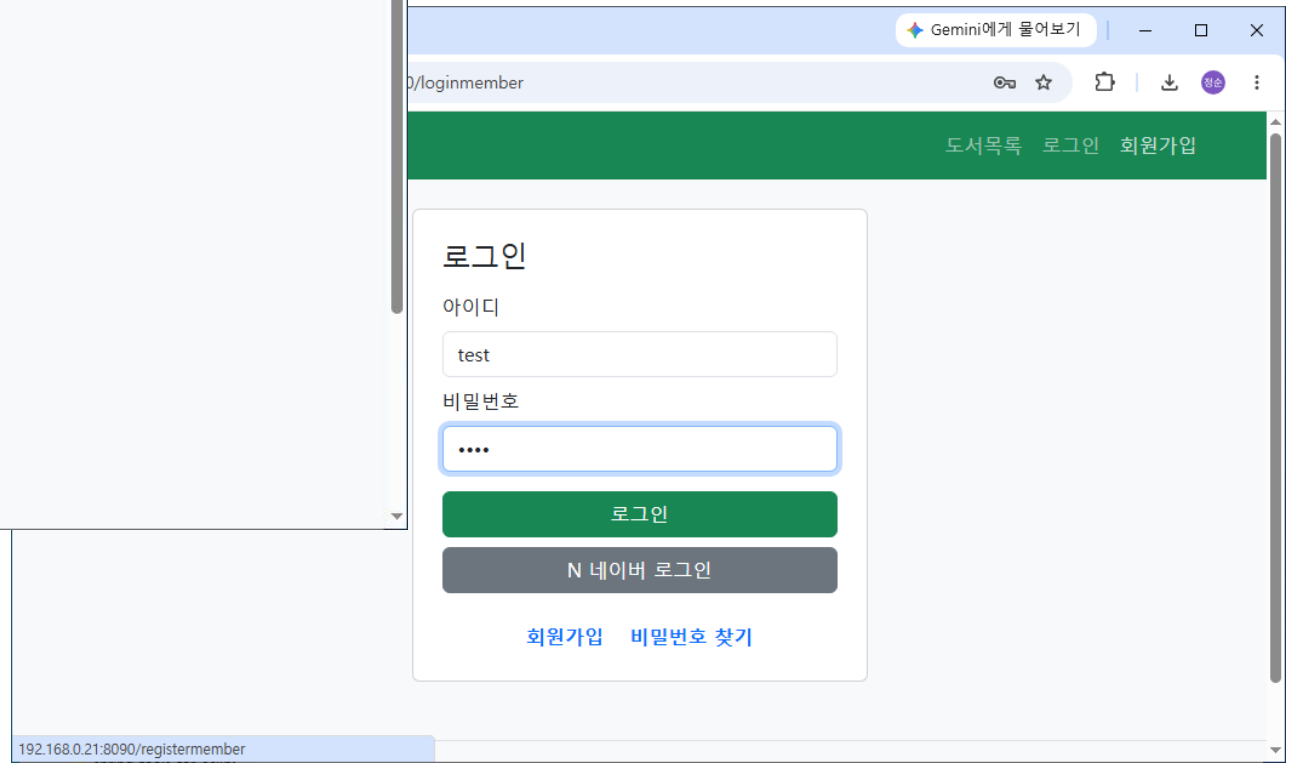
<div class="navbar-nav">
    <c:if test="${not empty sessionScope.loginMember}">
        <span class="nav-link fw-bold">${sessionScope.loginMember.m_name}님</span>
    </c:if>

    <a class="nav-link" href="booklist">도서목록</a>

    <c:choose>
        <c:when test="${empty sessionScope.loginMember}">
            <a class="nav-link" href="loginmember">로그인</a>
            <a class="nav-link" href="registermember">회원가입</a>
        </c:when>
        <c:otherwise>
            <a class="nav-link" href="bookrental">대출목록</a>
            <a class="nav-link" href="modifymember?m_no=${sessionScope.loginMember.m_no}">정보수정</a>
            <a class="nav-link" href="logoutmember">로그아웃</a>
        </c:otherwise>
    </c:choose>
</div>
```

■ 참고화면


AITECH 도서관
주의 요함
192.168.0.21:8090/booklist
AITECH 도서관
길동씨님 도서목록 대출목록 정보수정 로그아웃
도서명을 입력하세요
검색
도서 목록
김애란 소설
바깥은 여름 :김애란 소설
김애란 저자
작별하지 않는다
작별하지 않는다 :한강 장편 소설


Gemini에게 물어보기
/loginmember
도서목록 로그인 회원가입
로그인
아이디
test
비밀번호
로그인
N 네이버 로그인
회원가입 비밀번호 찾기
192.168.0.21:8090/registermember

도서 검색

BookController

- 검색시 검색어 유지

```
@RequestMapping("/searchbook")
public String searchbook(@RequestParam("searchTxt") String searchTxt, Model model) {
    model.addAttribute("books", bookDao.searchBook(searchTxt));
    model.addAttribute("searchTxt", searchTxt);
    return "book/book_list";
}
```

BookDao

```
public List<BookVo> searchBook(String searchTxt) {
    String sql = "SELECT * FROM book WHERE b_name LIKE ? ORDER BY b_no";
    return jdbcTemplate.query(sql, new BookRowMapper(), "%" + searchTxt + "%");
}
```

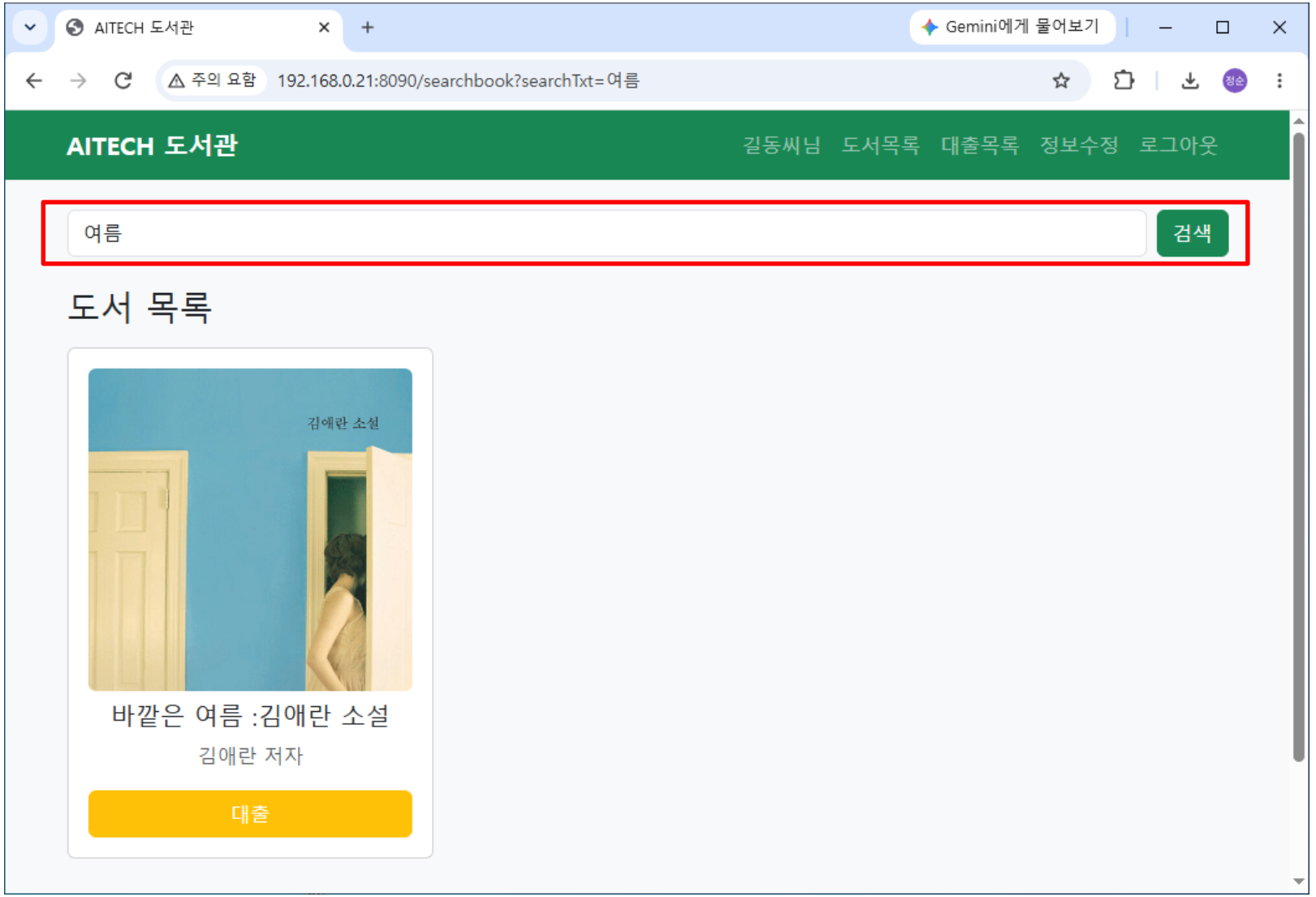
book_list.jsp

- <form> 태그 추가
- 검색어 URL 남기기 method GET방식 / 안 남기기 POST방식

```
<form method="get" action="searchbook" class="d-flex gap-2 mb-4">
    <input type="text" name="searchTxt" class="form-control"
        placeholder="도서명을 입력하세요" value="${searchTxt}">
    <button type="submit" class="btn btn-success flex-shrink-0">검색</button>
</form>
```

도서대출

■ 참고화면


AITECH 도서관
Gemini에게 물어보기
주의 요함
192.168.0.21:8090/searchbook?searchTxt=여름
AITECH 도서관
길동씨님 도서목록 대출목록 정보수정 로그아웃
여름
검색
도서 목록
김애란 소설
바깥은 여름 :김애란 소설
김애란 저자
대출

BookController

```
@RequestMapping("/rentalbook")
public String rentalbook(@RequestParam("b_no") int b_no, HttpSession session) {
    MemberVo loginMember = (MemberVo) session.getAttribute("loginMember");
    if (loginMember == null) {
        return "redirect:/loginmember";
    }

    BookVo book = bookDao.getDetailBook(b_no);
    if (book.getB_rental_able() != 1) {
        return "redirect:/bookdetail?b_no=" + b_no;
    }

    bookDao.insertRentalBook(b_no, loginMember.getM_no());
    bookDao.updateRentalAble(b_no, 0);
    return "redirect:/bookdetail?b_no=" + b_no;
}
```

BookDao

```
public void insertRentalBook(int b_no, int m_no) {
    String sql = "INSERT INTO rentalbook(b_no, m_no, rb_start_date,"
            + " rb_reg_date, rb_mod_date)"
            + " VALUES(?,?, now(), now(), now())";
    jdbcTemplate.update(sql, b_no, m_no);
}

public void updateRentalAble(int b_no, int b_rental_able) {
    String sql = "UPDATE book SET b_rental_able = ? WHERE b_no = ?";
    jdbcTemplate.update(sql, b_rental_able, b_no);
}
```

book_list.jsp

```
<div class="d-grid">
    <c:choose>
        <c:when test="${book.b_rental_able == 1}">
            <a href="rentalbook?b_no=${book.b_no}" class="btn btn-warning text-white"
                onclick="return confirm('[${book.b_name}] 대출하시겠습니까?');">대출</a>
        </c:when>
        <c:otherwise>
            <button type="button" class="btn btn-secondary" disabled>대출중</button>
        </c:otherwise>
    </c:choose>
</div>
```

book_detail.jsp

```
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<c:choose>
    <c:when test="${not empty book.b_img}">
        <img src="${book.b_img}" class="w-100 rounded"
            style="max-height: 420px; object-fit: contain;">
    </c:when>
    <c:otherwise>
        <div class="bg-light py-5 rounded" style="font-size:60px">책이미지</div>
    </c:otherwise>
</c:choose>

<h3 class="mb-0">${book.b_name}</h3>

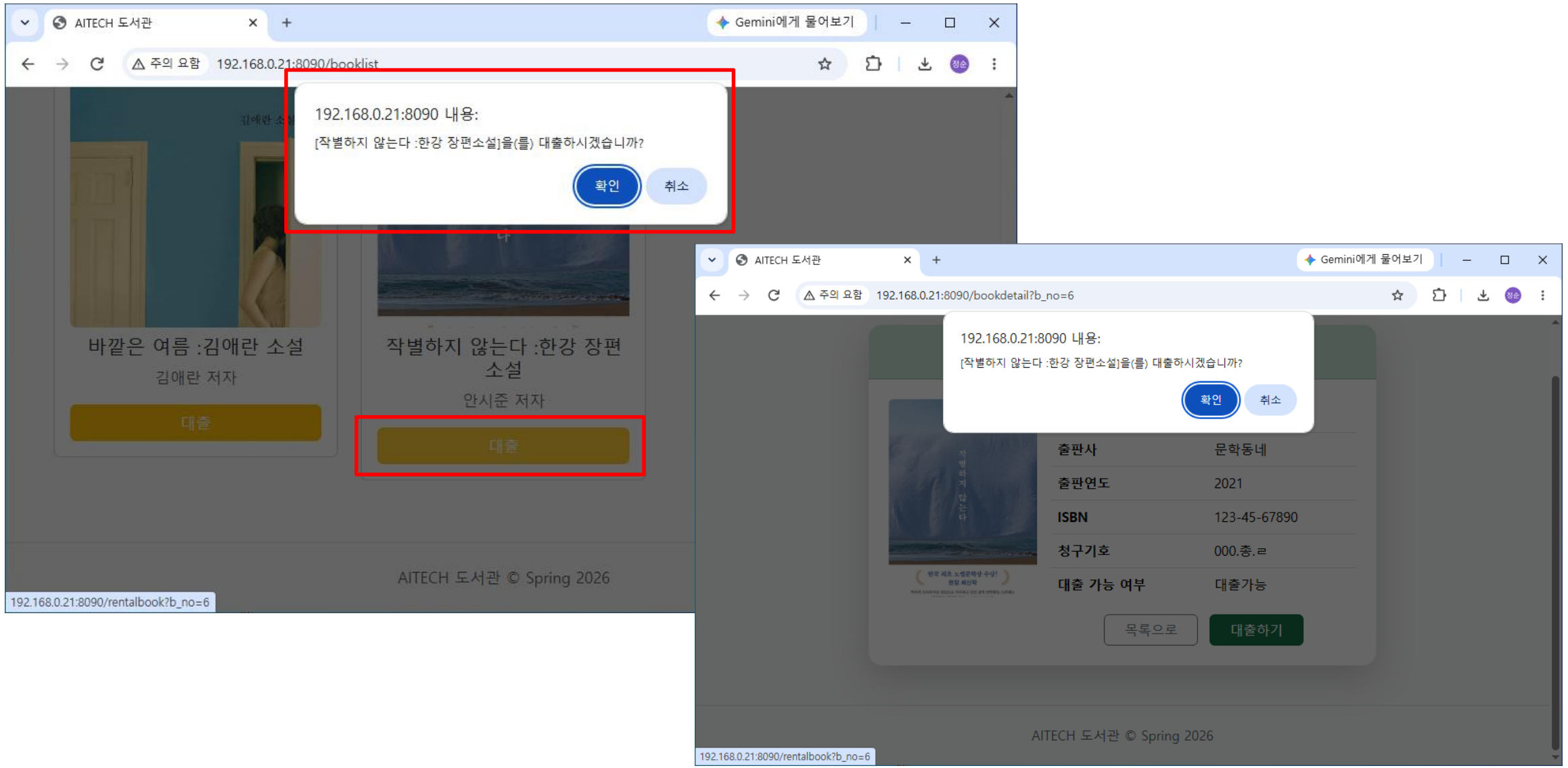
<td>${book.b_author}</td>
<td>${book.b_publisher}</td>
<td>${book.b_publish_year}</td>
<td>${book.b_isbn}</td>
<td>${book.b_call_number}</td>
```

book_detail.jsp

```
<td>
    <c:choose>
        <c:when test="${book.b_rental_able == 1}">대출가능</c:when>
        <c:otherwise>대출불가</c:otherwise>
    </c:choose>
</td>

<div class="text-center mt-4">
    <a href="booklist" class="btn btn-outline-secondary px-4 me-2">목록으로</a>
    <c:choose>
        <c:when test="${book.b_rental_able == 1}">
            <a href="rentalbook?b_no=${book.b_no}" class="btn btn-success px-4"
                onclick="return confirm('[${book.b_name}] 대출하시겠습니까?');">대출하기</a>
        </c:when>
        <c:otherwise>
            <button type="button" class="btn btn-secondary px-4" disabled>대출중</button>
        </c:otherwise>
    </c:choose>
</div>
```

참고화면


AITECH 도서관
Gemini에게 물어보기
주의 요함
192.168.0.21:8090/booklist
192.168.0.21:8090 내용:
[작별하지 않는다 :한강 장편소설]을(를) 대출하시겠습니까?
확인
취소
바깥은 여름 :김애란 소설
김애란 저자
대출
작별하지 않는다 :한강 장편 소설
안시준 저자
대출
AITECH 도서관 © Spring 2026
192.168.0.21:8090/rentalbook?b_no=6
AITECH 도서관
Gemini에게 물어보기
주의 요함
192.168.0.21:8090/bookdetail?b_no=6
192.168.0.21:8090 내용:
[작별하지 않는다 :한강 장편소설]을(를) 대출하시겠습니까?
확인
취소
출판사
문학동네
출판연도
2021
ISBN
123-45-67890
청구기호
000.총.ㄹ
대출 가능 여부
대출가능
목록으로
대출하기
AITECH 도서관 © Spring 2026
192.168.0.21:8090/rentalbook?b_no=6

대출 이력

BookController

```
@RequestMapping("/rentalbooklist")
public String rentalbooklist(HttpSession session, Model model) {
    MemberVo loginMember = (MemberVo) session.getAttribute("loginMember");
    if(loginMember ==  null) {
        return "redirect:/loginmember";
    }

    List<RentalBookVo> rentalbooks = bookDao.getMyRentalBook(loginMember.getM_no());
    model.addAttribute("rentalbooks", rentalbooks);
    return "book/book_rental";
}
```

BookDao

```
public List<RentalBookVo> getMyRentalBook(int m_no) {
    String sql = "SELECT rb.rb_no, rb.b_no,"
            + " DATE_FORMAT(rb.rb_start_date, '%Y-%m-%d') AS rb_start_date,"
            + " DATE_FORMAT(rb.rb_end_date, '%Y-%m-%d') AS rb_end_date,"
            + " b.b_name, b.b_call_number, b.b_img"
            + " FROM rentalbook rb JOIN book b ON rb.b_no = b.b_no"
            + " WHERE rb.m_no = ? ORDER BY rb.rb_no DESC";
    return jdbcTemplate.query(sql, new RentalBookVoRowMapper(), m_no);
}
```

RentalBookVo

```
public class RentalBookVo {
    private int rb_no;
    private int b_no;
    private String b_img;
    private String b_name;
    private String b_call_number;
    private String rb_start_date;
    private String rb_end_date;

    public int getRb_no() { return rb_no; }
    public void setRb_no(int rb_no) { this.rb_no = rb_no; }
}
```

RentalBookVoRowMapper

```
public class RentalBookDtoRowMapper implements RowMapper<RentalBookDto> {

    @Override
    public RentalBookDto mapRow(ResultSet rs, int rowNum) throws SQLException {
        RentalBookDto dto = new RentalBookDto();
        dto.setRb_no(rs.getInt("rb_no"));
        dto.setB_no(rs.getInt("b_no"));
        dto.setB_name(rs.getString("b_name"));
        dto.setB_call_number(rs.getString("b_call_number"));
        dto.setB_img(rs.getString("b_img"));
        dto.setRb_start_date(rs.getString("rb_start_date"));
        dto.setRb_end_date(rs.getString("rb_end_date"));
        return dto;
    }
}
```

book_rental.jsp

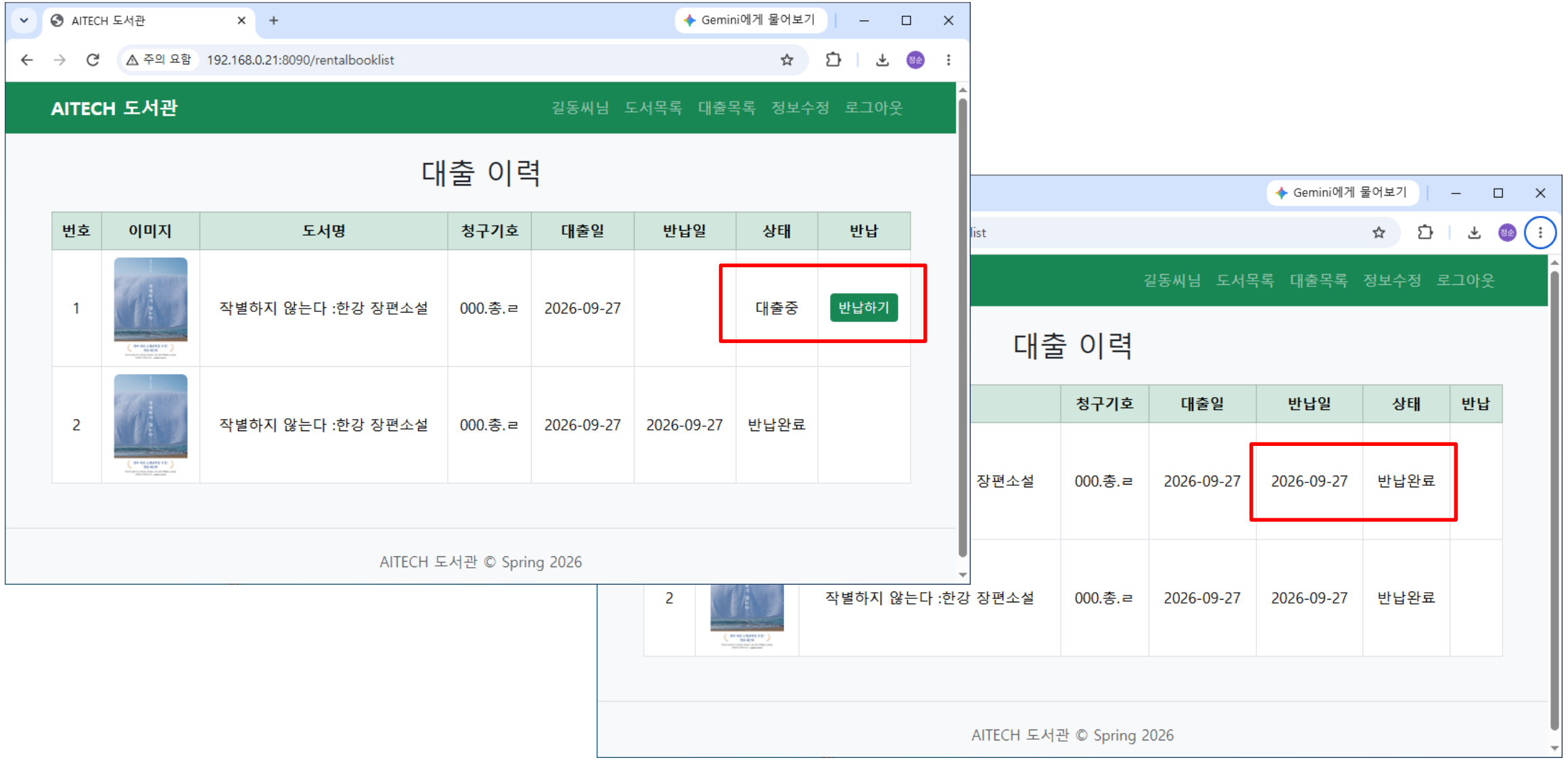
```
<%@ taglib prefix="c" uri="jakarta.tags.core" %>

<tbody>
    <c:forEach var="rentalbook" items="${rentalbooks}" varStatus="status">
        <tr>
            <td>${status.count}</td>
            <td>
                <c:choose>
                    <c:when test="${not empty rentalbook.b_img}">
                        <img src="${rentalbook.b_img}" class="rounded"
                            style="width: 80px; height: 110px; object-fit: cover;">
                    </c:when>
                    <c:otherwise>
                        <div class="bg-light rounded py-4 mx-auto" style="width: 80px;">책</div>
                    </c:otherwise>
                </c:choose>
            </td>
```

book_rental.jsp

```
                <td>${rentalbook.b_name}</td>
                <td>${rentalbook.b_call_number}</td>
                <td>${rentalbook.rb_start_date}</td>
                <td>${rentalbook.rb_end_date}</td>
                <td>
                    <c:choose>
                        <c:when test="${empty rentalbook.rb_end_date}">대출중</c:when>
                        <c:otherwise>반납완료</c:otherwise>
                    </c:choose>
                </td>
                <td>
                    <c:if test="${empty rentalbook.rb_end_date}">
                        <a href="returnbook?rb_no=${rentalbook.rb_no}&b_no=${rentalbook.b_no}"
                            class="btn btn-sm btn-success"
                            onclick="return confirm('[${rentalbook.b_name}] 반납하시겠습니까?');">반납하기</a>
                    </c:if>
                </td>
            </tr>
        </c:forEach>
    </tbody>
```

■ 대출이력 참고화면


AITECH 도서관
Gemini에게 물어보기
주의 요함
192.168.0.21:8090/rentalbooklist
AITECH 도서관
길동씨님 도서목록 대출목록 정보수정 로그아웃
대출 이력
번호 이미지 도서명 청구기호 대출일 반납일 상태 반납
1 작별하지 않는다 :한강 장편소설 000.총.ㄹ 2026-09-27 대출중 반납하기
2 작별하지 않는다 :한강 장편소설 000.총.ㄹ 2026-09-27 2026-09-27 반납완료
AITECH 도서관 © Spring 2026

도서 반납

BookController

```
@RequestMapping("/returnbook")
public String returnbook(@RequestParam("rb_no") int rb_no,
        @RequestParam("b_no") int b_no, HttpSession session) {
    if (session.getAttribute("loginMember") == null) {
        return "redirect:/loginmember";
    }
    bookDao.returnBook(rb_no);
    bookDao.updateRentalAble(b_no, 1);
    return "redirect:/bookrental";
}
```

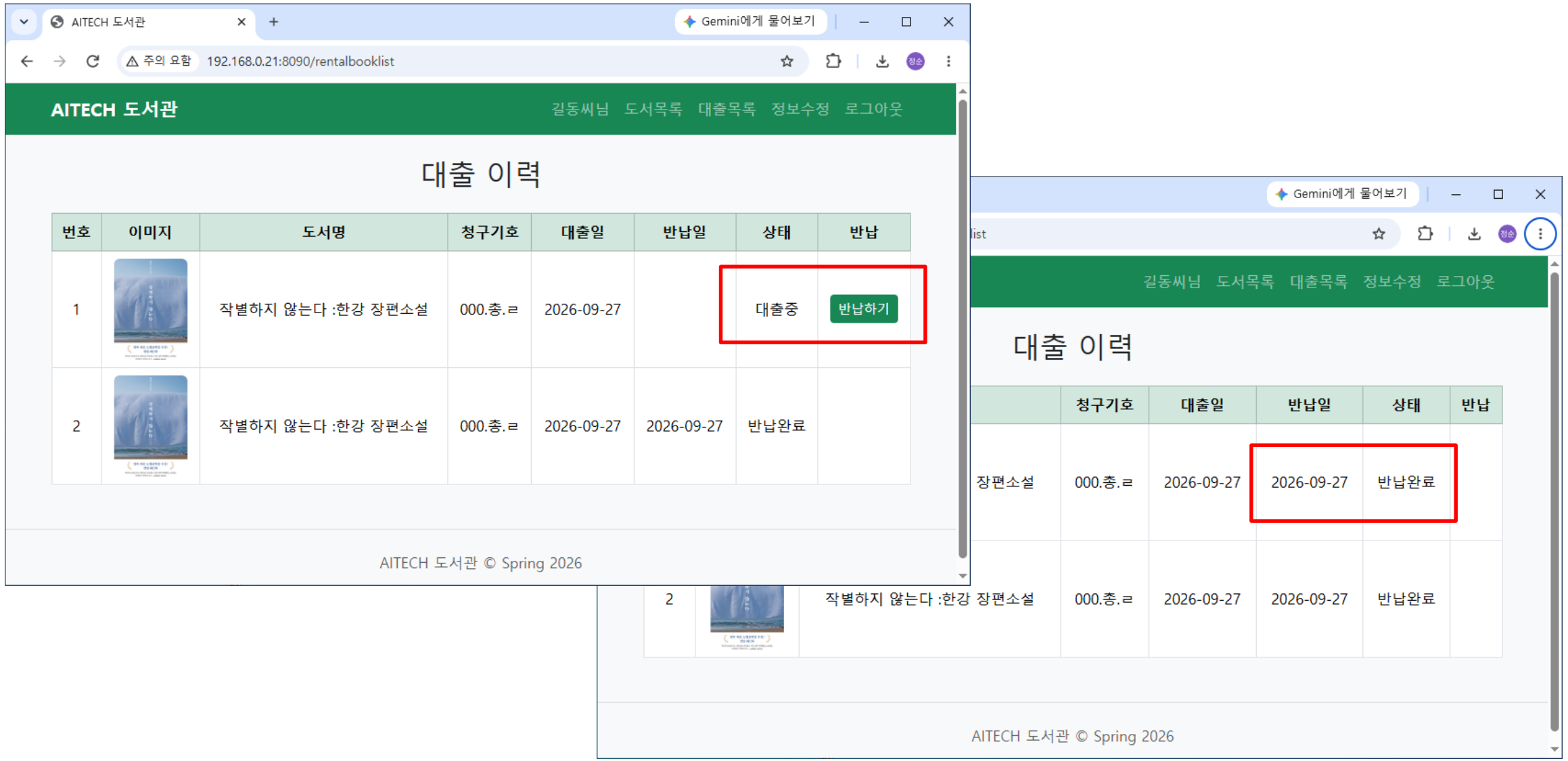
BookDao

```
public void returnBook(int rb_no) {
    String sql = "UPDATE rentalbook SET rb_end_date = now(),"
            + " rb_mod_date = now() WHERE rb_no = ?";
    jdbcTemplate.update(sql, rb_no);
}
```

book_rental.jsp

```
                <td>
                    <c:choose>
                        <c:when test="${empty rentalbook.rb_end_date}">대출중</c:when>
                        <c:otherwise>반납완료</c:otherwise>
                    </c:choose>
                </td>
                <td>
                    <c:if test="${empty rentalbook.rb_end_date}">
                        <a href="returnbook?rb_no=${rentalbook.rb_no}&b_no=${rentalbook.b_no}"
                            class="btn btn-sm btn-success"
                            onclick="return confirm('[${rentalbook.b_name}] 반납하시겠습니까?');">반납하기</a>
                    </c:if>
                </td>
            </tr>
        </c:forEach>
    </tbody>
```

■ 참고화면


AITECH 도서관
Gemini에게 물어보기
주의 요함
192.168.0.21:8090/rentalbooklist
AITECH 도서관
길동씨님 도서목록 대출목록 정보수정 로그아웃
대출 이력
번호 이미지 도서명 청구기호 대출일 반납일 상태 반납
1 작별하지 않는다 :한강 장편소설 000.총.ㄹ 2026-09-27 대출중 반납하기
2 작별하지 않는다 :한강 장편소설 000.총.ㄹ 2026-09-27 2026-09-27 반납완료
AITECH 도서관 © Spring 2026

Thank you!